Muhammad Daffa Husen

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Banda Aceh, Aceh, Indonesia

Computer Engineering graduate from Universitas Syiah Kuala with hands-on experience in Machine Learning, Web Development, IoT, and Embedded Systems. Experienced in developing and implementing technology-based projects, including BioCompost Buddy, an IoT-based smart composting system recognized as a Top 180 6th Innovillage 2025 project, as well as machine learning research involving SVM, IndoBERT, and IndoBERTweet. Completed an internship at PLN UID Aceh, contributing to digital content production and community empowerment initiatives. Proficient in developing web applications, IoT systems, and data-driven solutions, with strong problem-solving, adaptability, and teamwork skills.

Work Experiences

Universitas Syiah Kuala - **Banda Aceh** Aug 2025 - Jan 2026
Teaching Assistant - Embedded Systems Course

Spearheaded technical instruction and hardware integration for the Embedded Systems curriculum, bridging abstract microarchitectural theory with hands-on firmware engineering and practical embedded design.

- Successfully guided 20+ engineering students in architecting and deploying fully functional, hardware-validated embedded prototypes.
- Streamlined hardware debugging cycles during lab sessions by instituting structured diagnostic and troubleshooting protocols.
- Elevated overall project success rates and cohort technical proficiency through tailored code reviews and hardware interfacing workshops.

Universitas Syiah Kuala - **Banda Aceh** Jan 2025 - Jun 2025
Teaching Assistant - Software Engineering Course

Acted as a technical catalyst for the Software Engineering course, guiding students through the complete software development lifecycle to bridge complex theoretical concepts with real-world industry applications.

- Mentored 20+ students through the full Software Development Life Cycle (SDLC), driving end- to- end execution from requirements analysis to system implementation.
- Directed weekly hands-on laboratory sessions and evaluated 15+ technical assignments, delivering targeted feedback to accelerate practical proficiency.
- Co-developed 5+ advanced instructional modules with the lead instructor, optimizing course materials for modern software testing and maintenance.

PT PLN (Persero) Unit Induk Distribusi Aceh - **Banda Aceh** Dec 2024 - Jan 2025
Intern - Communication Division

Spearheaded critical digital initiatives within the Communication Division, heavily anchoring on the end-to-end development of the "Desa Berdaya PLN UID ACEH" web platform, while concurrently elevating corporate multimedia production and digital asset management.

- Architected and deployed the "Desa Berdaya PLN UID ACEH" website, driving end-to-end front-end development, UI/UX design, and content architecture to maximize community engagement.
- Engineered technical production for the "PodcaStroom" YouTube series, accelerating digital growth to secure 100 new subscribers within a single month alongside a consistent 200+ monthly views.
- Directed corporate media documentation, executing post- production for 500+ multimedia assets to establish a high- fidelity digital archive for strategic public relations and internal use.

Pekan Olahraga Nasional (PON) XXI Aceh-Sumatera Utara - **Banda Aceh** Sep 2024 - Oct 2024
Liaison Officer

Acted as the primary Liaison Officer for the Kalimantan Tengah official contingent during the PON XXI National Sports Week, managing critical communication and logistics to ensure seamless event coordination.

- Directed seamless communication between 5+ Central Kalimantan officials and the central organizing committee, achieving 100% schedule alignment with zero missed engagements.
- Coordinated daily logistical arrangements, schedules, and transportation across multiple venues, maintaining a 100% on-time arrival rate for the official delegation.
- Proactively resolved 10+ daily operational inquiries and specific participant needs, delivering high- quality assistance that directly contributed to the success of PON XXI.

Education

Universitas Syiah Kuala - **Banda Aceh** Aug 2022 - Aug 2026
Bachelor of Computer Engineering, 3.30/4.00

SMA Negeri 3 Banda Aceh - **Banda Aceh** Jul 2019 - Jun 2022
High School Diploma, Science Major, 90.35/100.00

Organizations & Leadership

Himpunan Mahasiswa Teknik Komputer (HIMATEKKOM) - **Banda Aceh** Mar 2025 - Dec 2025
Head of the Human Resources Development Division

Led the Human Resources Development Division of HIMATEKKOM, focusing on member development, organizational engagement, leadership, and team collaboration.

- Led the Human Resources Development Division, managing member development programs and internal organizational initiatives.
- Coordinated large-scale programs such as Computer Outbonding to strengthen teamwork, leadership, and member engagement.
- Built relationships with members, alumni, and external partners to support professional and organizational development.

Himpunan Mahasiswa Teknik Komputer (HIMATEKKOM) - **Banda Aceh** Mar 2024 - Dec 2024
Vice Chair of the Student Relations

Served as Vice Chair of Student Relations, strengthening collaboration between HIMATEKKOM, student organizations, alumni, and technology-sector partners.

- Built and maintained relationships with student associations, alumni, and technology-sector companies.
- Organized and supported programs that increased student participation in association activities.
- Coordinated student relations and cross-team collaboration for large-scale programs, including Computer Outbonding.

The IEEE International Conference on COSITE 2023 - **Banda Aceh** Aug 2023

Committee Member

Contributed as an organizing committee member for the 2nd IEEE COSITE 2023, an international conference dedicated to advancing research in computer systems, IT, and electrical engineering.

- Assisted in organizing conference logistics, venue preparations, and technical session workflows for local and international presenters.
- Managed on-site event execution, supporting audiovisual facilities and session scheduling during keynote and parallel presentation tracks.
- Coordinated attendee check-ins, distributed conference materials, and assisted speakers and delegates throughout the two-day event.

Bina Islami Aneuk Komputer (BINER 7.0) - **Banda Aceh** Oct 2023

Chairperson

Led the organizing committee of BINER 7.0, an educational initiative focused on exploring the integration of Society 5.0 with Islamic values.

- Led and coordinated the organizing team to execute BINER 7.0 in line with its vision and objectives.
- Managed event planning, budgeting, team coordination, and overall execution.
- Developed and implemented the event concept around the integration of Society 5.0 and Islamic values.

MPK-OSIS SMA Negeri 3 Banda Aceh - **Banda Aceh** Sep 2021 - Sep 2022

Head of Technology Information and Communication Division

Served as Head of the Information and Communication Technology Division at MPK- OSIS SMA Negeri 3 Banda Aceh, leading digital content, visual communication, and technology-related initiatives to strengthen the school's communication and media presence.

- Led the Information and Communication Technology Division, coordinating digital content and media initiatives.
- Designed engaging graphics, promotional materials, and visual content for school events and publications.
- Developed creative digital content, including educational posters, infographics, and video content, to increase audience engagement.

Awards

Innovillage 2025 Grant Recipient - Top 180 National Jan 2026

*Front-End Web Developer – PT PLN (Persero) UID Aceh Internship*Selected among the Top 180 funded projects from 995 nationwide proposals. Developed BioCompost Buddy, an IoT-based smart composting system with automated aeration, real-time sensor monitoring, and web dashboard; deployed in Gampong Cadek and trained 50 local farmers.

Finalist - UI/UX Design Competition Sep 2023

Informatics Festival, Universitas Syiah Kuala

Selected as a national finalist for LifeGen, a health-focused mobile application designed to track calorie intake, steps, burned calories, and food consumption. Applied user research, wireframing, prototyping, and usability testing using Figma.

Projects

Machine Learning and Transformer Based Sentiment Analysis of Roblox Game Reviews Oct 2025 - Aug 2026

Undergraduate Thesis Research – Universitas Syiah Kuala

Developed a sentiment analysis study to classify Indonesian-language Roblox user reviews into negative, neutral, and positive sentiments using machine learning and Transformer-based models.

- Scraped and preprocessed 50,000 Play Store reviews into 40,298 clean records, utilizing IndoBERT for automated sentiment labeling.
- Benchmarked classification performance between SVM (TF-IDF) and Transformer models (IndoBERT and IndoBERTweet).
- Achieved optimal performance with IndoBERT, reaching 88.91% accuracy and an 0.8178 macro F1-score.

BioCompost Buddy – Smart IoT Composting System Dec 2025 - Apr 2026

Innovillage 2025 - Telkom Indonesia

Selected as one of 180 funded projects out of 995 national proposals in Telkom Indonesia's Innovillage 2025, engineering an IoT-driven composting system to support rural food security in Gampong Cadek.

- Fabricated an IoT composting prototype featuring a motorized shredder, aeration, and ESP32 telemetry node.
- Built a real-time web dashboard with DHT22, soil moisture, and MQ-6 sensors to monitor fermentation.
- Led field deployment and trained 50 local farmers and village enterprise (BUMDes) members.

Website Desa Berdaya PLN UID Aceh Dec 2025 - Apr 2026

Front-End Web Developer – PT PLN (Persero) UID Aceh

Developed a responsive community empowerment website to showcase PLN-supported village programs, local Acehnese UMKM products, educational initiatives, and environmental sustainability projects across Aceh.

- Developed a responsive web platform showcasing UMKM products, village programs, maps, FAQs, and documentation.
- Implemented product filtering, media carousels, Google Maps, and responsive UI using HTML, CSS, JavaScript, Bootstrap 5, and Chart.js.
- Digitized community initiatives including GM English Course and Bank Sampah USK.

Skills

Programming Languages: Python, JavaScript, MicroPython, SQL
Machine Learning & Data: TensorFlow, Scikit-learn, Deep Learning, Computer Vision, Image Segmentation, NLP, Web Scraping, Image Processing
IoT & Embedded Systems: ESP32, Arduino, Sensor Integration, Hardware-Software Integration, Actuator Control, MQTT
Web & UI/UX: React.js, HTML5, CSS3, Figma, Git, Firebase
Soft Skills: Leadership, Project Management, Problem Solving, Cross-Functional Teamwork, Adaptability, Fast Learner